Bacterial Glycolate and Glyoxylate Assimilation: A Conserved Central-Carbon Module

A commissioned review-style synthesis

1. Executive Summary

Bacterial glycolate and glyoxylate assimilation is a compact, reusable central-carbon module that recovers two-carbon (C2) units and channels them back into core metabolism as the three-carbon intermediate **D-glycerate**. Functionally, the module is a linear cascade: 2-phosphoglycolate is dephosphorylated to glycolate, glycolate is oxidized to glyoxylate, two glyoxylate molecules are condensed (with loss of one CO_2) to tartronate semialdehyde, and tartronate semialdehyde is reduced to D-glycerate, which enters lower glycolysis/gluconeogenesis. This "glycerate pathway" is defined by three signature enzymes — a thiamine-diphosphate (ThDP)-dependent **glyoxylate carboligase (Gcl)**, an accessory **hydroxypyruvate isomerase (Hyi)**, and an NAD(H)-dependent **tartronate semialdehyde reductase (GlxR/TSR)** — typically encoded together in a dedicated operon, fed by an upstream, separately encoded **GlcDEF glycolate dehydrogenase** and an HAD-family **phosphoglycolate phosphatase (Gph)**.

The best-supported mechanistic model places the *obligatory* backbone at three steps — oxidation (glycolate → glyoxylate), carboligation (2 glyoxylate → tartronate semialdehyde), and reduction (tartronate semialdehyde → glycerate). Phosphoglycolate dephosphorylation is a *conditional feeder* whose flux depends on the carbon source (it is shared with photorespiration, the Calvin-cycle "phosphoglycolate salvage" of chemolithoautotrophs, and even DNA repair), and hydroxypyruvate isomerase is an *accessory* branch that connects hydroxypyruvate metabolism into the same tartronate-semialdehyde node. A crucial boundary distinction is that this glycerate pathway, although it consumes glyoxylate, is **not** the glyoxylate cycle (isocitrate lyase/malate synthase); the two are frequently co-induced on the same substrates yet are mechanistically and functionally separate anaplerotic modules.

The module is broadly conserved across Proteobacteria (with *Escherichia coli* and *Pseudomonas putida* KT2440 as the best-characterized exemplars) and Cyanobacteria, where it runs as one of three parallel glyoxylate-processing routes. The principal axis of lineage variation is the *form of the oxidation enzyme*: a three-subunit FAD/Fe-S GlcDEF complex in enteric and pseudomonad bacteria versus GlcD-type enzymes in cyanobacteria. The pathway is inducibly regulated by repressors (GlcC/GclR) responding to glycolate/glyoxylate, and its deregulation has been a recurrent target of laboratory evolution for growth on ethylene glycol and other one-/two-carbon feedstocks. This review synthesizes nine confirmed findings and 24 primary/secondary sources into a coherent mechanistic and evolutionary picture, and is explicit about where the evidence is strong (enzyme reactions, operon structure) versus where it is indirect or organism-specific (regulatory detail, complex stoichiometry, ancestral origins).

2. Definition and Biological Boundaries

What is included

The system under review is the **bacterial glycerate pathway for glycolate/glyoxylate assimilation**, comprising five biochemical steps and their catalysts:

Step	Reaction	Enzyme (family)	EC
1	2-phosphoglycolate → glycolate + P_i	Phosphoglycolate phosphatase, Gph (HAD superfamily)	3.1.3.18
2	glycolate → glyoxylate ($2e^-$ to quinone/ETC)	GlcDEF glycolate dehydrogenase (FAD/FAD/Fe-S)	1.1.99.14
3	2 glyoxylate → tartronate semialdehyde + CO_2	Glyoxylate carboligase, Gcl (ThDP/FAD, AHAS family)	4.1.1.47
4	hydroxypyruvate ⇌ tartronate semialdehyde	Hydroxypyruvate isomerase, Hyi	5.3.1.22
5	tartronate semialdehyde + NADH → D-glycerate	Tartronate semialdehyde reductase, GlxR/TSR (β-hydroxyacid DH)	1.1.1.60

The module boundary is deliberately drawn at the *conserved pathway segment* — the enzymatic conversions from 2-phosphoglycolate to D-glycerate — rather than at any single organism's gene locus. *P. putida* KT2440 supplies convenient local exemplars (Gph, GlcDEF, Gcl, Hyi, GlxR), but the same reaction logic recurs in *E. coli*, cyanobacteria, and many other heterotrophs and autotrophs.

Also within scope, as the transport boundary, is **glycolate uptake**: in *E. coli* the 2-hydroxymonocarboxylate permease **GlcA (yghK)** imports glycolate, with the L-lactate permease **LldP** providing a redundant route (PMID: 11283302).

What should be treated separately (adjacent, often-confused processes)

- **The glyoxylate cycle** (isocitrate lyase AceA + malate synthase GlcB/AceB). This shares the glyoxylate metabolite node and is frequently **co-induced** with the glycerate pathway on glycolate/ethylene-glycol/glyoxylate substrates, but it is a distinct anaplerotic bypass of the TCA cycle that condenses glyoxylate with acetyl-CoA to make malate. Confusingly, in *E. coli* the malate synthase G gene (*glcB*) sits within the same *glc* operon as glycolate oxidase (PMID: 8606183), and in *P. putida* Gcl, GlcB and AceA are induced together during ethylene-glycol growth (PMID: 23023748). These are two separate glyoxylate-consuming modules and must not be conflated.
- **Plant/algal photorespiration (the C2 cycle)**. Photorespiration also begins with 2-phosphoglycolate produced by Rubisco's oxygenase activity, and it shares the phosphoglycolate phosphatase step. But the eukaryotic C2 cycle routes glyoxylate through glycine/serine aminotransferases and spans chloroplast, peroxisome and mitochondrion. The bacterial glycerate pathway is a cytoplasmic, aminotransferase-independent alternative. Cyanobacteria notably contain *both* a plant-like C2 cycle and the bacterial glycerate pathway (PMID: 18957552).
- **DNA-repair-derived 2-phosphoglycolate metabolism**. In heterotrophs, 2PG also arises from repair of 3′-phosphoglycolate DNA ends (e.g., after bleomycin damage), and Gph salvages it — a source entirely unrelated to carbon assimilation (PMID: 23195894).

Competing definitions

The literature does not agree on a single name. In photosynthetic contexts the first step is "photorespiratory 2PG metabolism"; Claassens et al. proposed the broader, photosynthesis-neutral term **"phosphoglycolate salvage"** to cover non-photosynthetic Calvin-cycle autotrophs (PMID: 32820073). The C2-recovery segment from glyoxylate to glycerate is variously called the **"glycerate pathway"** or the **"glyoxylate carboligase pathway."** These terms overlap but emphasize different boundaries; this review treats the glycerate pathway (steps 1–5 above) as the unit of analysis while flagging where "phosphoglycolate salvage" and "glyoxylate cycle" are distinct.

---

3. Mechanistic Overview

Best current model of the sequence of events

```
DNA repair / photorespiration / Calvin-cycle salvage
                     │  (2-phosphoglycolate)
                     ▼
 [1] Gph (HAD)   2-phosphoglycolate ──► glycolate + Pi     (CONDITIONAL feeder)
                     │
 glycolate uptake: GlcA / LldP permeases
                     ▼
 [2] GlcDEF      glycolate ──► glyoxylate                  (OBLIGATORY)
     (FAD/FAD/Fe-S, 2e⁻ → quinone/ETC)
                     ▼
 [3] Gcl (ThDP)  2 glyoxylate ──► tartronate semialdehyde + CO2   (OBLIGATORY)
                     ▲
 [4] Hyi         hydroxypyruvate ⇌ tartronate semialdehyde  (ACCESSORY feed)
                     ▼
 [5] GlxR (NADH) tartronate semialdehyde ──► D-glycerate    (OBLIGATORY)
                     ▼
           D-glycerate ──► 2-/3-phosphoglycerate ──► central carbon metabolism
```

The load-bearing logic is **oxidation → carboligation → reduction**. Glycolate is first oxidized to glyoxylate (step 2). Two glyoxylate molecules are then condensed by glyoxylate carboligase, which decarboxylates one to form the three-carbon aldehyde tartronate semialdehyde (2-hydroxy-3-oxopropanoate; step 3). Finally, tartronate semialdehyde is reduced to D-glycerate (step 5), which is phosphorylated to enter glycolysis/gluconeogenesis. Two molecules of C2 glycolate thus yield one C3 glycerate plus one CO_2, conserving three of four carbons.

Which steps are obligatory, conditional, accessory

- **Obligatory backbone (steps 2, 3, 5):** oxidation, carboligation, reduction. In *P. putida*, co-expression of only *gcl* and *glxR* (plus native oxidation capacity) was sufficient to confer growth on ethylene glycol, identifying **gcl + glxR as the minimal functional core** of the glycerate branch (PMID: 29885475).

- **Conditional feeder (step 1):** phosphoglycolate dephosphorylation is only required when 2PG is the entry metabolite (photorespiration, Calvin-cycle autotrophy, DNA repair). When cells grow on glycolate, glyoxylate, or ethylene glycol directly, step 1 is bypassed.
- **Accessory (step 4):** hydroxypyruvate isomerase connects hydroxypyruvate metabolism to the same tartronate-semialdehyde node but is not required for glycolate assimilation per se. Its equilibrium favors tartronate semialdehyde (K = [TSA]/[hydroxypyruvate] ≈ 2.5 at pH 7.1; PMID: 7448201).

Molecular assemblies carrying out each step

- **Step 1 — Gph:** an HAD-superfamily hydrolase with a characteristic α-helical cap domain and Zn/Mg dependence (structural work in *Shigella*; PMID: 19153451).
- **Step 2 — GlcDEF:** an obligate three-subunit membrane-associated dehydrogenase; GlcD and GlcE are flavoprotein subunits, GlcF is an iron-sulfur protein (PMID: 8606183).
- **Step 3 — Gcl:** a homo-oligomeric ThDP/FAD enzyme of the acetohydroxyacid synthase (AHAS)/pyruvate oxidase family (PMID: 8440684).
- **Step 4 — Hyi:** a cofactor-independent homodimer (~58 kDa in *E. coli*) (PMID: 10561547).
- **Step 5 — GlxR/TSR:** an NAD(H)-dependent β-hydroxyacid dehydrogenase (β-hydroxyisobutyrate dehydrogenase family) (PMID: 10978349).

---

4. Major Molecular Players and Active Assemblies

4.1 GlcDEF glycolate dehydrogenase — the obligate three-subunit oxidation complex

The *E. coli glc* locus (min 64.5) is the founding characterization of this complex. Pellicer et al. showed the locus contains a divergently transcribed regulator (*glcC*) and the structural genes *glcD, glcE, glcF, glcG, glcB*. The proteins from *glcD* and *glcE* resemble flavoenzymes, and GlcF resembles iron-sulfur proteins — establishing the FAD/FAD/Fe-S subunit composition. Critically, **insertional inactivation of glcD, glcE, OR glcF each abolished glycolate oxidase activity**, demonstrating that all three are obligatory subunits of a single complex; by contrast, disruption of *glcG* had no measurable effect (PMID: 8606183). The permease boundary was defined by Núñez et al.: **GlcA (yghK)**, located 350 bp downstream of *glcB*, is a hydrophobic 2-hydroxymonocarboxylate permease highly similar to L-lactate permease (LldP), and LldP provides a redundant glycolate-uptake route (PMID: 11283302). This is a quinone-linked dehydrogenase (feeding electrons to the respiratory chain), not an O_2-dependent oxidase in the peroxisomal sense — an important mechanistic distinction from eukaryotic glycolate oxidase.

Verbatim evidence:

> "The proteins encoded from glcD and glcE displayed similarity to several flavoenzymes, the one from glcF was found to be similar to iron-sulfur proteins" (PMID: 8606183)
>
> "The insertional mutation ... in either glcD, glcE, or glcF abolished glycolate oxidase activity, indicating that presumably these proteins are subunits of this enzyme" (PMID: 8606183)

4.2 Glyoxylate carboligase (Gcl) — the ThDP-dependent condensation engine

Gcl (EC 4.1.1.47) catalyzes the committed, carbon-conserving step of the pathway: condensation of two glyoxylate molecules to tartronic semialdehyde. Chang, Wang & Cronan cloned and sequenced the *E. coli* enzyme (593 aa) and placed it firmly in the **AHAS/pyruvate-oxidase family** (30% similarity to acetohydroxyacid synthases, 26% to pyruvate oxidase), with binding sites for **thiamine pyrophosphate, FAD, and a quinone**. They also showed that the downstream *orf258* is not a subunit, since its disruption did not affect growth on glyoxylate or glycolate (PMID: 8440684). The ThDP cofactor is mechanistically essential: like other ThDP enzymes, Gcl forms a covalent enamine/carbanion intermediate on the thiazolium C2 that enables acyl-anion (umpolung) chemistry for C–C bond formation (PMID: 19476484).

> "Glyoxylate carboligase (Gcl) (EC 4.1.1.47) of Escherichia coli catalyzes the condensation of two molecules of glyoxylate to give tartronic semialdehyde, a key intermediate in glyoxylate catabolism" (PMID: 8440684)

4.3 Hydroxypyruvate isomerase (Hyi) — the accessory feed

Ashiuchi & Misono demonstrated that *E. coli hyi* (orf b0508/gip), located immediately downstream of *gcl*, encodes a **58-kDa homodimer that exclusively interconverts hydroxypyruvate and tartronate semialdehyde**, requires no cofactor, has Km(hydroxypyruvate) = 12.5 mM, and an optimum pH of 6.8–7.2 (PMID: 10561547). The reaction's equilibrium was earlier quantified in *Bacillus fastidiosus*: K = [TSA]/[hydroxypyruvate] = 2.5 at pH 7.1 (PMID: 7448201). Because both Gcl and Hyi produce the same intermediate (tartronate semialdehyde), Hyi effectively bridges hydroxypyruvate metabolism (e.g., from serine catabolism) into the GlxR reduction step.

4.4 Tartronate semialdehyde reductase (GlxR/TSR) — the terminal reduction to glycerate

Njau, Herndon & Hawes identified the *E. coli* β-hydroxyacid dehydrogenase that **specifically catalyzes NAD^+-dependent oxidation of D-glycerate and NADH-dependent reduction of tartronate semialdehyde**, defining it as tartronate semialdehyde reductase. Decisively for pathway organization, its coding region is **directly preceded by hydroxypyruvate isomerase (hyi) and glyoxylate carboligase (gcl)**, forming an operon "clearly designed for D-glycerate biosynthesis from tartronate semialdehyde." The *Haemophilus influenzae* paralogue prefers 4-carbon β-hydroxyacids, illustrating family-level substrate divergence (PMID: 10978349).

> "The purified recombinant protein very specifically catalyzed the NAD(+)-dependent oxidation of d-glycerate and the NADH-dependent reduction of tartronate semialdehyde, identifying this protein as a tartronate semialdehyde reductase" (PMID: 10978349)

4.5 Phosphoglycolate phosphatase (Gph) — the conditional entry hydrolase

Gph is an HAD-superfamily phosphatase. In *E. coli* it metabolizes 2-phosphoglycolate formed during DNA repair of 3′-phosphoglycolate ends and physically interacts with GAPDH, implicating it in a DNA-integrity protein complex (PMID: 23195894; PMID: 25603270). The *Shigella* orthologue's structure revealed the diagnostic HAD α-helical cap domain and metal (Zn/Mg) dependence (PMID: 19153451). Phylogenetically, phosphoglycolate phosphatase is unusual among photorespiratory enzymes in appearing to originate from **Archaea** (PMID: 26931168).

4.6 Operon organization (P. putida KT2440 exemplar)

Franden et al. showed that the four genes flanking *gcl* — *hyi, glxR, ttuD, pykF* — are transcribed as a single **operon**, and that co-expression of only *gcl* and *glxR* enabled ethylene-glycol growth, while full-operon expression improved utilization; separately, overexpression of the *glcDEF* glycolate-oxidase operon relieved the glycolate/glycolaldehyde bottleneck (PMID: 29885475). Thus the module is typically split into **two operons**: a glcDEF oxidation operon and a gcl-hyi-glxR(-ttuD-pykF) glycerate-branch operon.

---

5. Evolutionary and Cell-Biological Variation

5.1 Variation in the oxidation enzyme across lineages

The principal axis of lineage variation is the *form of the glycolate-oxidation catalyst*:

Lineage	Oxidation enzyme	Electron acceptor	Notes
Enterobacteria (*E. coli*)	GlcDEF three-subunit complex (FAD/FAD/Fe-S)	quinone/ETC	Obligate 3-subunit; GlcA/LldP uptake
Pseudomonads (*P. putida*)	GlcDEF (glcDEF operon)	quinone/ETC	Overexpression relieves glycolate bottleneck
Cyanobacteria (*Synechocystis*)	Two GlcD isoforms (GlcD1/GlcD2)	membrane ETC	ΔglcD1/ΔglcD2 cannot grow at low CO_2
Plants/algae (comparison)	Peroxisomal glycolate oxidase (O_2-dependent, H_2O_2-producing)	O_2	Different enzyme class; photorespiration

5.2 Cyanobacteria run three parallel glyoxylate routes

Eisenhut et al. established that *Synechocystis* sp. PCC 6803 possesses **three routes for 2PG/glyoxylate metabolism**: (i) the plant-like photorespiratory C2 cycle, (ii) the **bacterial glycerate pathway** (gcl/glxR-type), and (iii) complete glyoxylate decarboxylation via oxalate. All three converge on glyoxylate, which is synthesized by two GlcD isoforms. A ΔglcD1/ΔglcD2 double mutant could not grow at low CO_2, and a triple mutant blocking all three routes required high CO_2 despite an active CO_2-concentrating mechanism (PMID: 18957552). This makes the glycerate pathway one of several redundant options in oxygenic phototrophs and demonstrates convergence on glyoxylate as a metabolic hub.

> "this cyanobacterium also possesses the bacterial glycerate pathway and is able to completely decarboxylate glyoxylate via oxalate" (PMID: 18957552)

5.3 Physiological-state variation: induction and laboratory evolution

The pathway is inducible, not constitutive. The *E. coli glc* structural genes show induced expression controlled by the divergently transcribed regulator **GlcC**, with no single polycistronic transcript for the whole locus (PMID: 8606183). In *P. putida*, the repressor **GclR** represses the glyoxylate carboligase pathway as part of a larger purine/allantoin metabolic context, and **loss-of-function gclR mutations were central to evolving growth on ethylene glycol** (PMID: 31166064). This regulatory theme recurs in metabolic-engineering studies: the module has been recruited to consume ethylene glycol, 1,4-butanediol, and other PET-derived and one-/two-carbon feedstocks (PMID: 29885475; PMID: 23023748; PMID: 37067433; PMID: 37918546).

5.4 Compartmentalization

In bacteria the entire module is cytoplasmic (with GlcDEF membrane-associated). In eukaryotic photorespiration the homologous chemistry is distributed across chloroplast, peroxisome and mitochondrion — a fundamental cell-biological difference. Engineering the bacterial Gcl+Hyi into plant peroxisomes to short-circuit photorespiration produced stress phenotypes and did not cleanly recapitulate the bacterial route, underscoring that compartmental context matters (PMID: 22104170).

---

6. Conservation, Origin, and Constraints

6.1 Evolutionary origin

The individual enzyme families are ancient and were assembled from pre-existing folds rather than invented de novo:

- **Gcl** belongs to the ThDP-dependent AHAS/pyruvate-oxidase superfamily, one of the oldest enzyme lineages in central carbon and amino-acid metabolism (PMID: 8440684).
- **GlxR/TSR** belongs to the β-hydroxyacid (β-hydroxyisobutyrate) dehydrogenase family, whose members have diversified in substrate preference across bacteria (PMID: 10978349).

- **Gph** is an HAD-superfamily hydrolase, and phylogenetic analysis of photorespiratory enzymes places phosphoglycolate phosphatase's origin uniquely in **Archaea** — implying a deep, possibly pre-cyanobacterial provenance for the dephosphorylation step distinct from the other pathway enzymes (PMID: 26931168).
- The **endosymbiotic transfer** of photorespiratory enzymes from cyanobacteria to plastid-bearing eukaryotes, with subsequent losses and replacements (e.g., glycerate 3-kinase lost, serine:glyoxylate aminotransferase replaced by the α-proteobacterial version), shows a reticulate history in which the bacterial glycerate-pathway components are the plausible ancestral core in the phototroph lineage (PMID: 23551942; PMID: 26931168).

Best representatives for the ancestral role: within the expanded β-hydroxyacid dehydrogenase family, the *E. coli* enzyme with strict D-glycerate/tartronate-semialdehyde specificity is the best model for the glycerate-pathway role, whereas the *H. influenzae* paralogue (4-carbon substrate preference) illustrates a derived function (PMID: 10978349). For the oxidation step, GlcD-type single subunits (cyanobacteria) versus the GlcDEF three-subunit complex (Proteobacteria) frame the question of which arrangement is ancestral — currently unresolved.

6.2 Ordering and mutual-exclusivity constraints

- **Strict order:** oxidation must precede carboligation (Gcl requires glyoxylate), and carboligation must precede reduction (GlxR requires tartronate semialdehyde). This is dictated by substrate identity: phosphoglycolate phosphatase produces glycolate, the substrate oxidized by GlcDEF; GlcDEF supplies the glyoxylate used by glyoxylate carboligase; Gcl produces tartronate semialdehyde, the substrate reduced by GlxR.
- **Convergence at glyoxylate:** glyoxylate is a hub consumed by at least three competing fates — the glycerate pathway (Gcl), the glyoxylate cycle (malate synthase), and (in cyanobacteria) oxidative decarboxylation. Flux partitioning at this node is a key control point.
- **Redox coupling:** step 5 requires NADH; the reductive terminus balances the oxidative step 2, but because step 2 feeds the respiratory chain (quinone) rather than producing NADH directly, the redox economy depends on the electron-transport context.
- **Carbon accounting:** the carboligation step loses one CO_2 per two glyoxylates, so the pathway is inherently carbon-lossy (75% carbon retention from glyoxylate to glycerate); this constrains its yield relative to routes that conserve all carbon.

6.3 Failure modes

Metabolic-engineering studies reveal the module's fragilities: accumulation of cytotoxic aldehyde intermediates (glycolaldehyde, glyoxylate) and imbalance between acetyl-CoA and glycolyl-CoA pools limit flux; overexpression of glcDEF and balancing of the two operons were required to relieve bottlenecks (PMID: 29885475; PMID: 37067433). In plants, forcing bacterial Gcl into peroxisomes without functional Hyi produced necrotic lesions — a deleterious short-circuit of the photorespiratory nitrogen cycle (PMID: 22104170).

---

7. Controversies and Open Questions

Strongly supported claims (direct biochemistry/genetics): - The reactions and cofactors of Gcl (ThDP/FAD), Hyi (cofactor-independent), and GlxR (NAD(H)) are established by purified-enzyme assays (PMID: 8440684; PMID: 10561547; PMID: 10978349). - The obligate three-subunit nature of GlcDEF is established by single-gene knockouts each abolishing activity (PMID: 8606183). - The gcl-hyi-glxR operon organization is documented in both *E. coli* and *P. putida* (PMID: 10978349; PMID: 29885475).

Areas of uncertainty, indirect evidence, or cross-organism mixing: - **GlcDEF quaternary structure and stoichiometry.** Subunit identity is genetically established, but the precise stoichiometry, membrane topology, and electron-transfer path of the assembled complex are not fully resolved; no high-resolution structure of the intact bacterial complex is cited here. - **GlcD single- vs. multi-subunit forms.** Whether cyanobacterial GlcD acts alone or with unidentified partners, and which arrangement is ancestral, is unsettled (PMID: 18957552). - **Regulatory logic.** GlcC (activator/repressor role) and GclR repression are documented, but the effector molecules, operator sites, and cross-talk with purine/allantoin regulation are only partially defined (PMID: 8606183; PMID: 31166064). - **In vivo flux partitioning at glyoxylate** between the glycerate pathway and the glyoxylate cycle under different carbon sources is inferred from co-induction rather than measured by flux analysis (PMID: 23023748). - **Gph provenance.** The Archaeal origin of phosphoglycolate phosphatase is a phylogenetic inference and the functional significance of its distinct history is unclear (PMID: 26931168). - **2PG as a signal.** Whether 2PG is a bona fide internal signal of inorganic-carbon deprivation in cyanobacteria remains only partially supported in vivo (PMID: 25297829).

Most important open questions: (1) a structural model of the assembled GlcDEF complex and its electron-transfer chemistry; (2) quantitative flux control at the glyoxylate node in vivo; (3) the molecular effectors and operator architecture of GlcC/GclR; (4) whether the single-subunit GlcD or the three-subunit GlcDEF is the ancestral oxidation arrangement.

---

8. Limitations and Knowledge Gaps

This synthesis rests almost entirely on a handful of model organisms — *E. coli*, *P. putida* KT2440, and *Synechocystis* — and on individual purified-enzyme studies that are decades old for some steps (e.g., Hyi kinetics from *Bacillus fastidiosus*, 1980). Generalization to the broader bacterial world should be cautious: enzyme families are conserved, but subunit arrangements, operon structures, and regulatory wiring vary and are not comprehensively sampled. Several mechanistic claims (complex stoichiometry, regulatory effectors, in vivo flux) are indirect. No new experimental data were generated in this investigation; it is a literature-grounded synthesis of nine confirmed findings across 24 sources.

---

9. Proposed Follow-up Experiments / Actions

1. **Structural biology of GlcDEF.** Determine a cryo-EM or crystal structure of the assembled three-subunit complex to resolve stoichiometry, FAD/Fe-S arrangement, and quinone-binding site — directly addressing the largest mechanistic gap.
2. **^{13}C metabolic flux analysis** on cells grown on glycolate/glyoxylate/ethylene glycol to quantify partitioning between the glycerate pathway and the glyoxylate cycle at the glyoxylate node.
3. **Regulatory mapping** of GlcC and GclR: DNA-footprinting/ChIP to define operator sites and identification of the small-molecule effectors (glycolate, glyoxylate, tartronate semialdehyde?).
4. **Comparative genomics/phylogenetics** across bacterial phyla to test whether single-subunit GlcD or three-subunit GlcDEF is ancestral, and to map the distribution of the gcl-hyi-glxR operon.

5. **Reconstitution and kinetic characterization** of the module as a minimal in vitro cascade (Gph → GlcDEF → Gcl → GlxR) to measure rate-limiting steps and intermediate toxicity thresholds relevant to feedstock engineering.
6. **Aldehyde-toxicity mitigation** engineering: co-express aldehyde detoxification with balanced glcDEF/gcl-operon expression to raise ethylene-glycol/1,4-butanediol conversion titers.

---

10. Key References

PMID	Contribution
8606183	*glc* locus; obligate three-subunit GlcDEF (FAD/FAD/Fe-S); inducible regulation by GlcC; glcB co-localization
11283302	GlcA glycolate permease; LldP redundancy (transport boundary)
8440684	Gcl reaction, ThDP/FAD cofactors, AHAS/pyruvate-oxidase family
10561547	Hyi: cofactor-independent hydroxypyruvate/TSA isomerase, downstream of gcl
7448201	Hyi equilibrium constant (K=2.5) in *B. fastidiosus*
10978349	GlxR/TSR: NAD(H)-dependent, D-glycerate product; gcl-hyi-glxR operon
29885475	*P. putida* gcl-hyi-glxR-ttuD-pykF operon; gcl+glxR minimal core
31166064	GclR repressor; role in ethylene-glycol laboratory evolution
23023748	Co-induction of Gcl, GlcB, AceA on ethylene glycol (module vs. glyoxylate cycle)
18957552	Three parallel cyanobacterial glyoxylate routes; GlcD isoforms
23195894	Gph in DNA-repair 2PG metabolism; GAPDH interaction
25603270	GAPDH–Gph link to DNA repair
19153451	*Shigella* PGPase structure: HAD cap domain, metal dependence
26931168	Photorespiratory enzyme phylogeny; PGPase Archaeal origin
32820073	"Phosphoglycolate salvage" broader definition
23551942	Endosymbiotic origin/loss/replacement of photorespiratory enzymes
22104170	Bacterial Gcl/Hyi engineered into plant peroxisomes
19476484	ThDP enzyme reaction mechanisms
25297829	2PG as putative low-carbon signal in cyanobacteria
37067433	1,4-butanediol degradation via GCL pathway in *P. putida*
37918546	Ethylene glycol as next-generation feedstock
41316996	Glycolic acid production from ethylene glycol (*C. glutamicum*)
41038355	Glycolic acid production in *K. pneumoniae*

PMID	Contribution
30870683	Photorespiration/allantoin context (adjacent process)

Prepared as a commissioned review synthesis. Claims are anchored to the cited primary and secondary literature; uncertainty is flagged where evidence is indirect or organism-specific. The bacterial glycerate pathway (glycolate/glyoxylate → D-glycerate) should be understood as a conserved, inducible central-carbon salvage module, kept mechanistically distinct from the glyoxylate cycle and from eukaryotic photorespiration despite shared metabolite nodes.